ROBAYET ROWSHAN SIAM

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PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate at United International University with a CGPA of 3.82/4.00. Experienced in developing AI-powered applications, implementing algorithms and building software systems through hands-on academic projects. Proficient in Python, Java, C/C++, data processing and database management, with practical experience in machine learning, reinforcement learning and explainable AI. Interested in research, software engineering, and developing intelligent technology-driven solutions.

EDUCATION

BSc in Computer Science and Engineering 2023 – Present
United International University, Dhaka, Bangladesh
CGPA: 3.82/4.00
Expected Graduation: February 2027

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, PHP
Web Technologies: HTML, CSS, JavaScript
Database: MySQL
AI Libraries & Frameworks: Scikit-learn, Librosa, Whisper, DeepFace, NumPy, Pandas, Matplotlib
Development Tools: Git, GitHub, VS Code, Kaggle, Google Colab, Overleaf, Figma

RELEVANT COURSEWORK

Core & Architecture: Data Structures & Algorithms, System Design & Analysis, Software Engineering
Web & Interaction: Web Programming, Human-Computer Interaction
Data & Intelligent Systems: Database Management Systems, Artificial Intelligence, Machine Learning, Data Mining, Digital Image Processing

ACADEMIC PROJECTS

AI Confidence Coach – Multimodal ML System for Confidence Measure *GitHub*
Python, Scikit-learn, Librosa, Whisper, SHAP, Stable-Baselines3 (PPO), DeepFace, FastAPI

- Built a system that predicts speaking confidence (0–100) from 34 audio features (Librosa, Whisper) fused with DeepFace facial emotion scores and explains each score in plain language using SHAP.
- Created a 550-clip dataset from RAVDESS, CREMA-D, and TED-LIUM rated by 3 annotators (ICC = 0.737); trained a Random Forest regressor (MAE = 7.53 on a 0–100 scale).
- Used SHAP to find that pause ratio, speech rate and pitch variation drove scores and that Whisper strips filler words, making the filler feature unreliable.
- Designed a PPO agent that selects among 8 coaching exercises; evaluated on 50 held-out clips and real voice recordings and analyzed score compression and RL policy collapse with planned fixes.

MeramotHub – Local Service Booking Platform *GitHub-Frontend | GitHub-Backend*
React, Tailwind CSS, Node.js, Supabase (PostgreSQL), JWT

- Built a four-role platform (Customer, Service Provider, Area Manager, Admin) with role-based dashboards, location-based worker search, ratings and job tracking.
- Designed an escrow payment flow (hold, completion confirmation, commission deduction, release) with MFS payments.
- Implemented complaint handling that freezes payment and assigns an Area Manager, plus a Groq-powered AI chatbot for service recommendations.

Algorithm Quest – Multiplayer Algorithm Learning Game *GitHub*
Java, JavaFX, Scene Builder, Multithreading, Socket Programming

- Built a story-driven game with three levels (BFS, Bubble Sort, Linear Search) that visualize traversal, swaps and sequential search in JavaFX.
- Implemented real-time two-player BFS mode using a client-server socket architecture and multithreading for synchronized timers and animations.

RideSafety – Ride-Sharing Safety App for Women *Figma*
User research, Usability testing, Figma

- Turned survey and interview findings into design requirements and a high-fidelity prototype with emergency SOS (offline SMS fallback), live location sharing, route deviation alerts and anonymous incident reporting.
- Contributed to task-based usability evaluation, with 90–100% task completion on most core tasks.